Experiment 10

Atomic Spectra

Objective	Materials Used
To observe and measure emission spectral lines and to determine the Rydberg constant.	Gas tube and power supply, Pasco Atomic Spectra equipment, Capstone Interface

Theory

A hot gaseous element emits light of different frequencies. The frequencies of the light are characteristic of the particular element. Balmer in 1885 developed an empirical relationship that correctly described the spectrum of hydrogen.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

Where $R_H = 1.097 \times 10^7 m^{-1}$ is the Rydberg constant, λ is the wavelength, and n is an integer value greater than 2. Neils Bohr, in 1913, was able to derive the Balmer relationship by assuming that energy and angular momentum were quantized.

Procedure

1. Equipment 1

   (a) If the spectrophotometer needs to be assembled, refer to the assembly instructions in the Spectrophotometer manual. NOTE: The diffraction grating should be on the side of the grating mount that is toward the collimating lens and the light source. The grating side of the plate glass must be on the 0-180-degree line on the degree table.

   (b) Put the spectrophotometer on two rod stands so you can adjust the height to match the height of the various light sources.

   (c) Plug the Rotary Motion Sensor and the High Sensitivity Light Sensor into the interface.

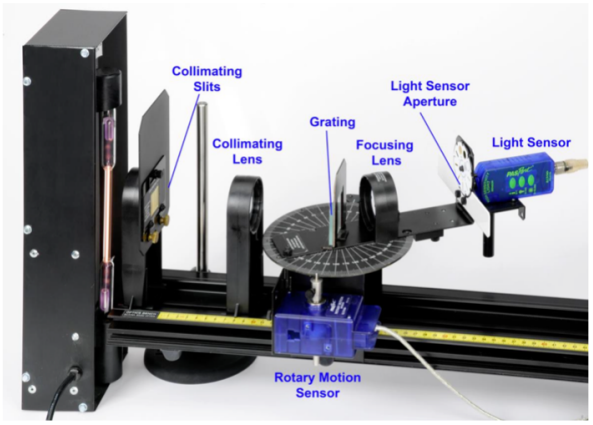

Figure 1: Setup

Figure 10.1: Reference Atomic Spectra

2. Equipment 2

   (a) Carefully insert the helium spectral tube into the Spectral Tube Power Supply and plug in the Power Supply.

   (b) The Collimating Slit must be at the focal point of the first lens and the Aperture Disk must be at the focal point of the second lens. Move the spectroscopy table back to the end of the track so it is out of the way. Move the Collimating Lens (see Figure 3) at least 12 cm from the slit. Have someone with 20/20 vision (corrected by glasses is fine) look through the lens at the slit. Move the lens toward the slit until it first comes into sharp focus. The slit should be about 10 cm from the lens. Now move the spectroscopy table as close to the Collimating Lens as possible. Set the Focusing Lens 10 cm from the Sensor Mask. We will adjust this more exactly in step 3.

   (c) Looking from the back side of the Collimating Slit, center the 4 slit in the hole and tighten the screw that holds the Collimating Slits in place.

   (d) Turn on the helium light source. Move the track up or down on the rod stands so the Collimating Slit is toward the brightest part of the helium tube. Swing the Light Sensor Arm out of the way so you can look through the grating along the Optics Track. Move the light source so that the light that you see through the slit is as bright as possible. Be careful not to move the track after it is set correctly.

   (e) Swing the Light Sensor Arm so that the Sensor Mask is along the track and the bright 0th order spectrum (undeviated) is on the Aperture Disk. Adjust the Focusing Lens so the image on the Aperture Disk is as sharp as possible. The system is now well collimated.

3. Software

   (a) Open Pasco Capstone. Make sure your sensor interface is connected and turned on.

   (b) The Rotary Motion pin rotates 60 times for each revolution of the spectrophotometer disk. To account for this, create a calculation in the Capstone Calculator:

   $$\text{Table Angle} = [Angle, ChP2(rad)]/60\text{with units of radians (rad)}$$

   (c) In Capstone, create a graph and select Light Intensity (% of scale max) on the vertical axis and Table Angle on the horizontal axis.

   (d) Set the common sample rate for the sensors on 25 Hz.

4. Experiment 1: Helium Calibration Use the yellow line in the helium (He) spectrum to determine the separation, d, of the lines on the grating. The yellow line of helium has a wavelength of 587.46 nm.

   (a) The following settings are suggested to use with all experiments:

      i. Light Sensor Button: 0-1

      ii. Collimating Slit: 4

      iii. Light Sensor Slit: 2

   (b) Start the scan at the central maximum and scan very slowly towards the first side maximum. The angles may all be negative, depending on how you set up the spectrometer. This just indicates direction and can be considered when doing calculations. (Be smarter than your tools!)

   (c) Using the Smart Tool and the Delta Tool on the graph, measure the angle from the central maximum to the first side maximum ($m = 1$) for the yellow line.

   (d) Using the wavelength of the yellow helium line, 587.46 nm, calculate the grating line separation using the diffraction equation. The accepted value is printed on the box for the diffraction grating. This is the value of d that you will use for the rest of this experiment.

5. Experiment 2: Balmer Series

   (a) Replace the helium tube with the hydrogen tube. CAUTION: These tubes can be very hot! Adjust the height of the spectrophotometer to match the center of the hydrogen tube. Start the scan at the central maximum and scan very slowly towards the first side maximum. You should measure at least three different colors before getting to $m = 2$.

   (b) Plot $\frac{1}{\lambda}$ vs $\frac{1}{n^2}$ to find R_H. Compare your measured value with the accepted value.

6. Experiment 3: Spectrometer

   This experiment can be performed rather quickly using a spectrometer (you may have used in chemistry to examine oscillating atoms). These devices are calibrated using known wavelengths, just like we did in this lab.

   (a) Make a new calculation in Pasco Capstone that relates your measured diffraction grating value (d) and the angle of the sensor to calculate the wavelength. Assume that we will only see light from $m = 1$.

   (b) Create a plot of the Intensity of the light vs wavelength (as calculated above). What do the peaks of this graph tell you?

Conclusion

Report your measured slit value, d, and compare it to the manufacture value. Report your value of R_H based on your plot. Compare it with the accepted value. Comment on the spectrometer graph you made based on your measured values.